PRODUCT INVENTORY MANAGEMENT SYSTEM

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Team Size : 4
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Role 1 – Project Overview

Our project is Product Inventory Management System. It is a backend application developed using Node.js, Express.js, and MongoDB.

This system helps users manage product inventory easily. Users can create an account, log in securely, and perform operations such as adding, viewing, updating, and deleting products.

Objectives: • Secure user authentication • Manage product inventory • Each user can access only their own products • Build scalable backend system

Role 2 – Technologies and Architecture

Technologies Used: • Node.js – Backend runtime • Express.js – API framework • MongoDB – Database • JWT – Authentication • bcrypt – Password hashing

Database Structure: Users Collection – name, email, password
Products Collection – name, price, quantity, description, userId

Relationship: One User → Many Products

Role 3 – System Features and User Flow

Core Features: • User authentication • Product CRUD operations • Private data access • Product search

User Flow: 1. User registers or logs in 2. System generates JWT token 3. Token is sent with requests 4. Middleware verifies token 5. Controller performs operations 6. MongoDB stores/retrieves data 7. API sends response

Role 4 – Implementation and Testing

Project Setup Steps: 1. Created Node.js project 2. Installed required packages 3. Connected MongoDB database 4. Created models and controllers 5. Created routes and middleware 6. Tested APIs using Thunder Client

API Testing

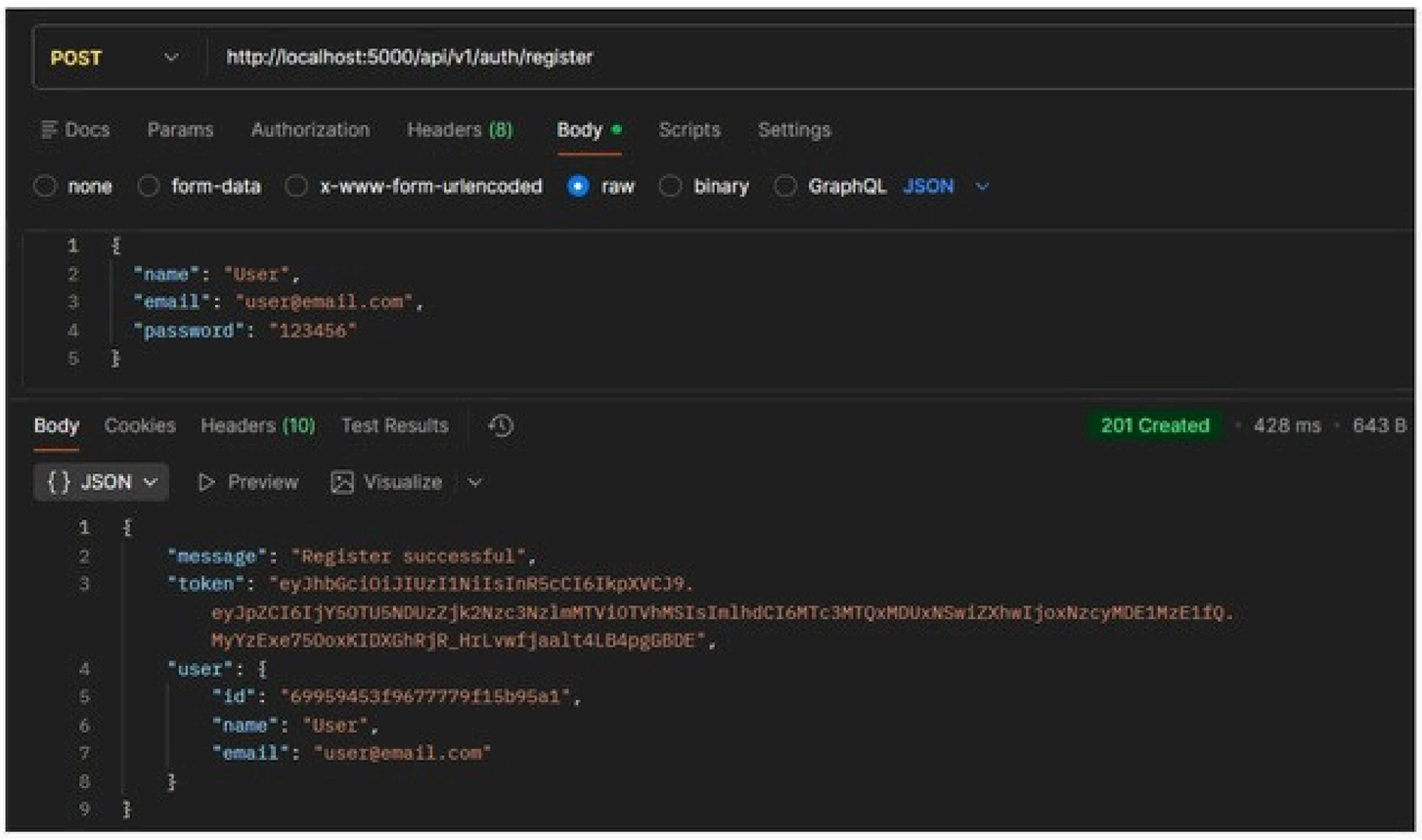
The following screenshots show testing of APIs using Thunder Client.

API Testing :

Use Thunder Client to test APIs:

Test APIs:

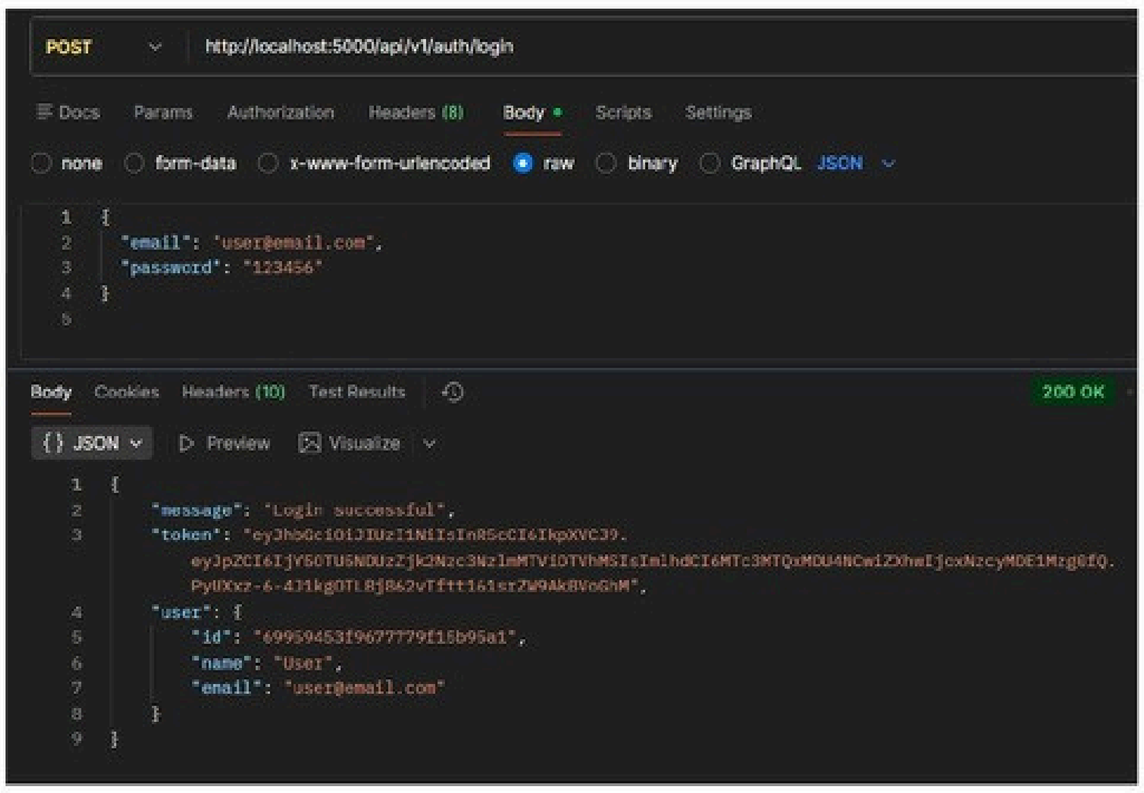
Register User:



Register:

The register functionality allows new users to create an account by submitting their details, which are validated and securely stored with a hashed password. It ensures unique user creation before granting access to the system.

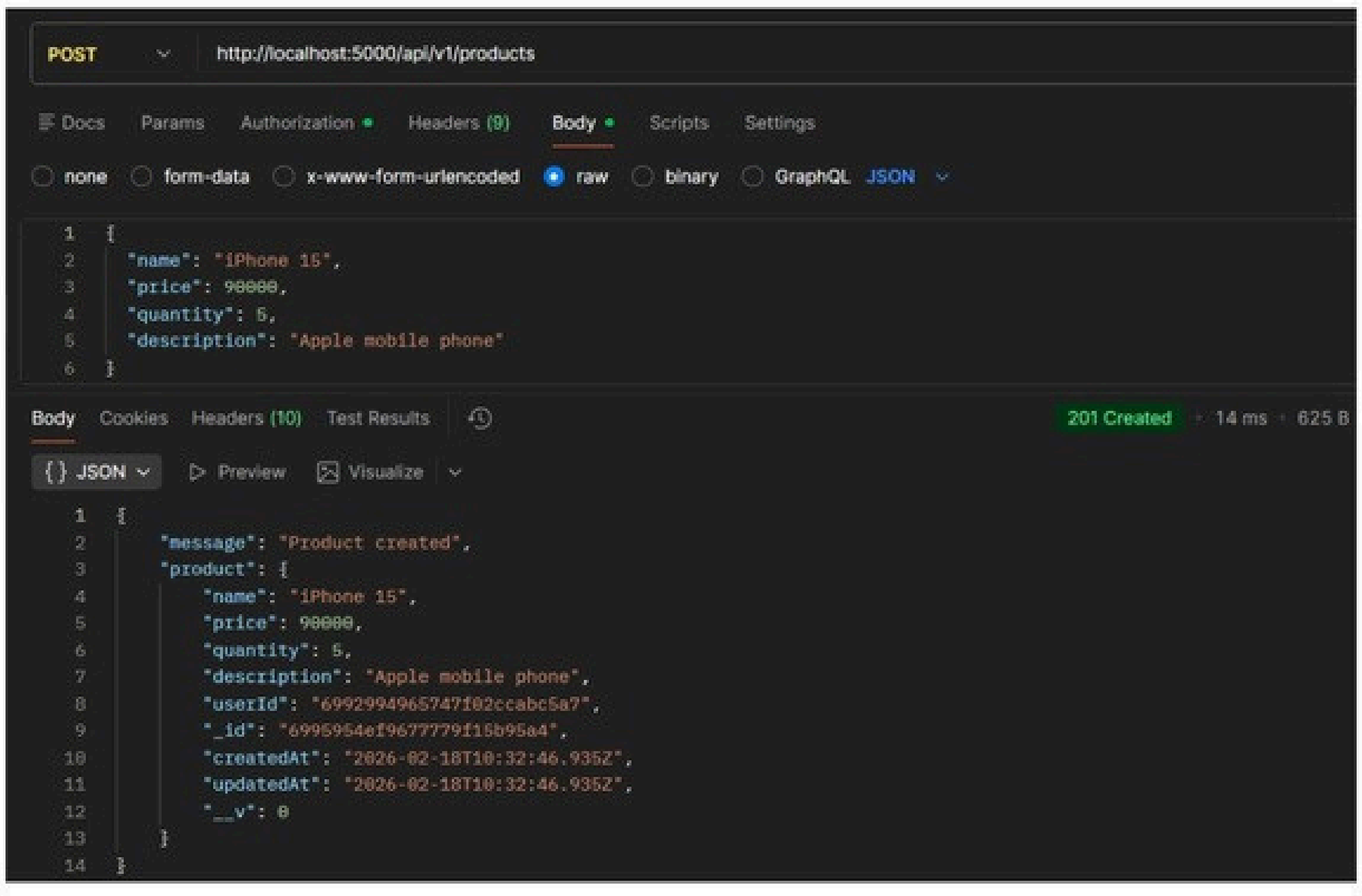
Login User



Login:

The login functionality verifies user credentials and generates a JWT token upon successful authentication. This token is used to access protected routes securely.

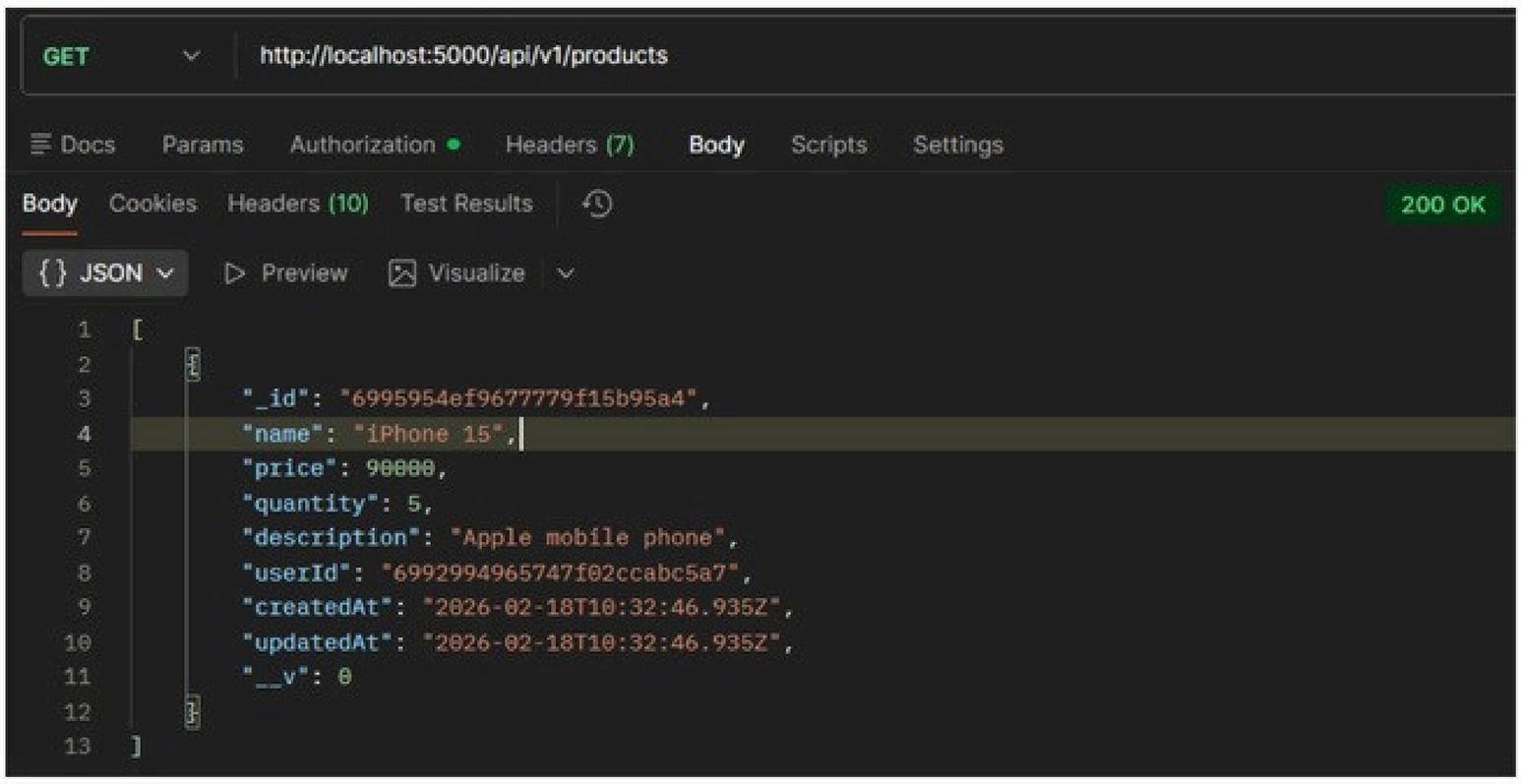
Create Product



Create Product:

Allows an authenticated user to add a new product to their inventory by submitting product details. The product is saved with the user's ID to maintain ownership.

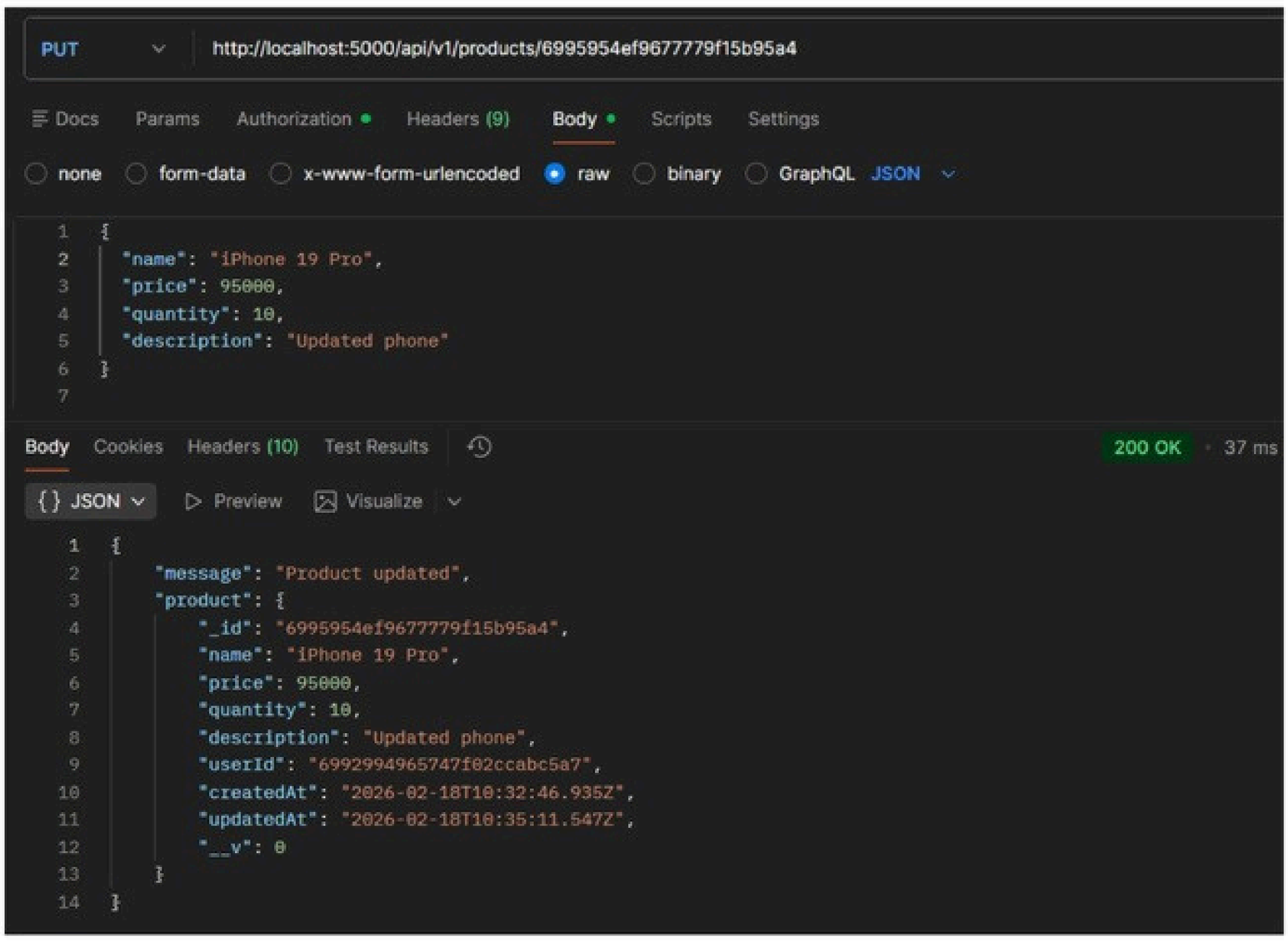
Get all Products



Get all Products:

Fetches all products that belong to the logged-in user. It ensures users can view only their own inventory data.

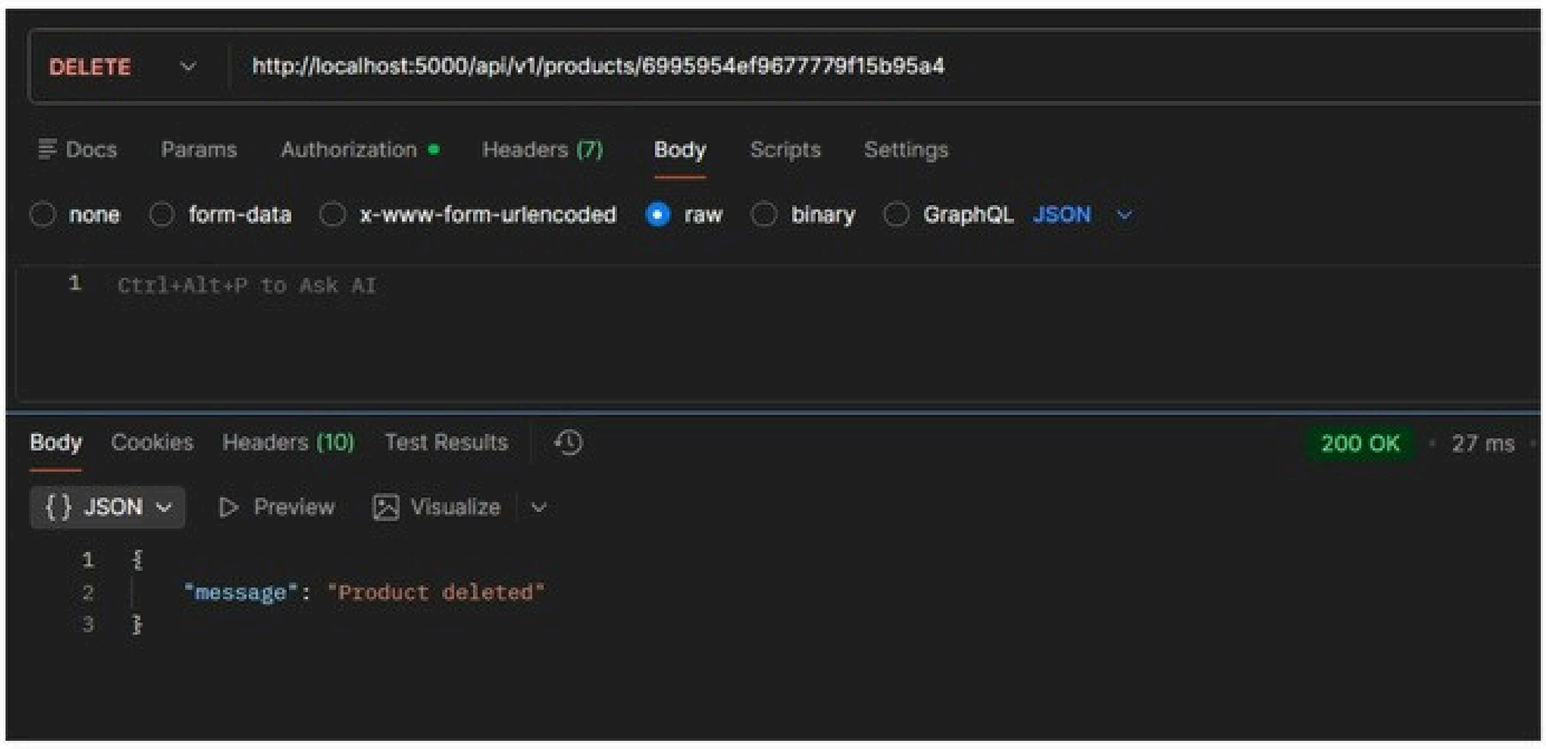
Update Product



Update Product:

Enables users to modify existing product details such as name, price, or quantity. The system verifies ownership before allowing any updates.

Delete Product



Delete Product:

Allows users to permanently remove a product from their inventory. The operation is protected to ensure only the product owner can delete it.